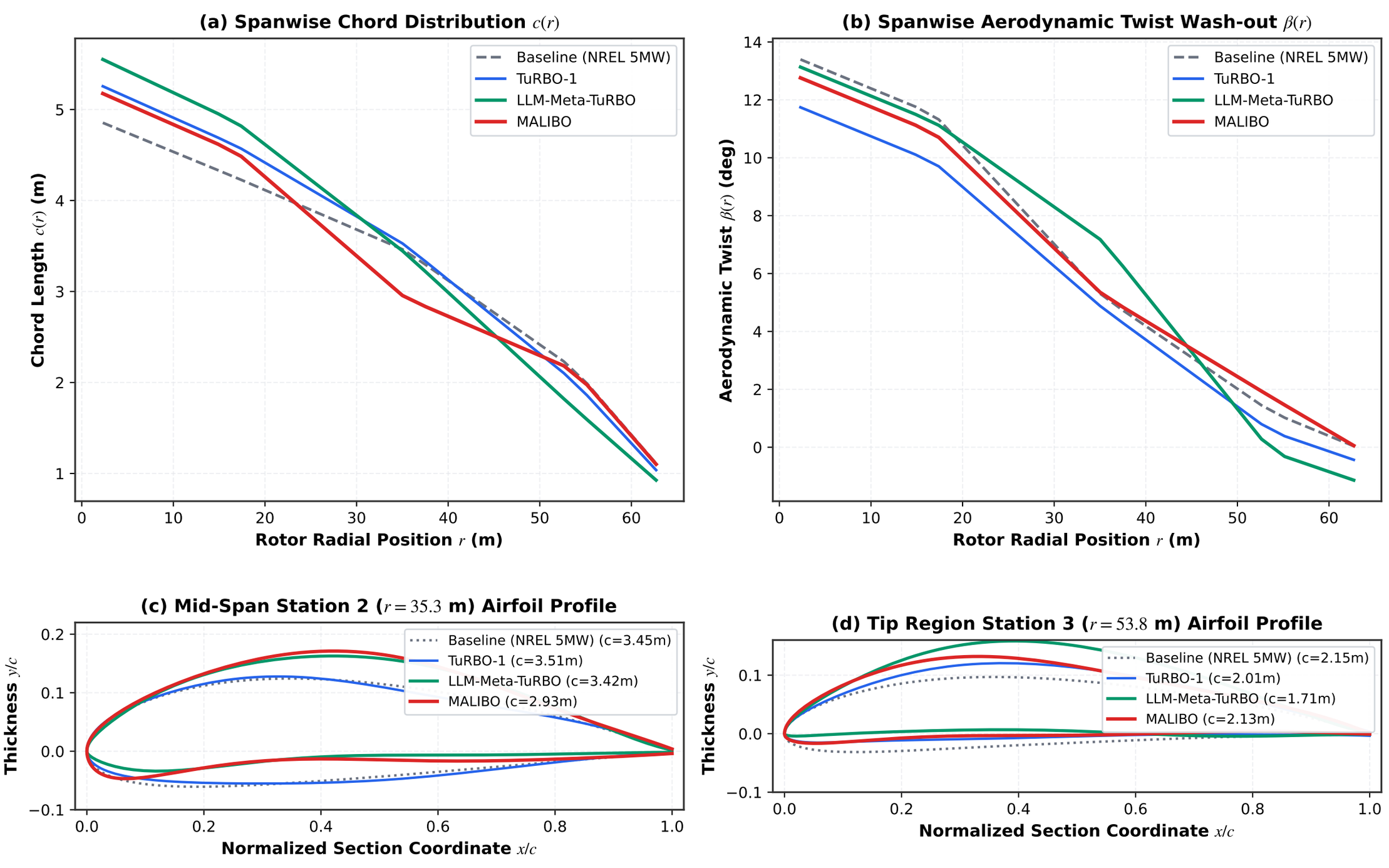
3D Wind Turbine Blade Structural & Aerodynamic Lofting Comparison
(Representative Median Configurations after 300 Function Evaluations)
(a) Spanwise Chord Distribution $c(r)$
Chord Length $c(r)$ (m)
Rotor Radial Position r (m)
Baseline (NREL 5MW)
TuRBO-1
LLM-Meta-TuRBO
MALIBO
(b) Spanwise Aerodynamic Twist Wash-out $\beta(r)$
Aerodynamic Twist $\beta(r)$ (deg)
Rotor Radial Position r (m)
(c) Mid-Span Station 2 ($r = 35.3$ m) Airfoil Profile
Thickness y/c
Normalized Section Coordinate x/c
Baseline (NREL 5MW) (c=3.45m)
TuRBO-1 (c=3.51m)
LLM-Meta-TuRBO (c=3.42m)
MALIBO (c=2.93m)
(d) Tip Region Station 3 ($r = 53.8$ m) Airfoil Profile
Thickness y/c
Normalized Section Coordinate x/c
Baseline (NREL 5MW) (c=2.15m)
TuRBO-1 (c=2.01m)
LLM-Meta-TuRBO (c=1.71m)
MALIBO (c=2.13m)